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Course Code: CSE 260

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Section: 27A

[Assignment-1]

[Ans to the question no 1]

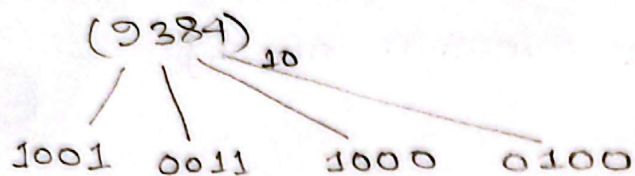
$$(9384)_{10} = (\quad)_2$$

2		9384	
2		4692	— 0
2		2346	— 0
2		1173	— 0
2		586	— 1
2		293	— 0
2		146	— 1
2		73	— 0
2		36	— 1
2		18	— 0
2		9	— 0
2		4	— 1
2		2	— 0
2		1	— 0
		0	— 1

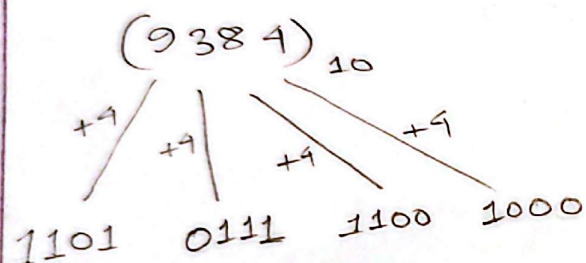


So,

$$(9384)_{10} = (10010010101000)_2$$



$$(9384)_{10} = (1001 \ 0011 \ 1000 \ 0100)_{BCD}$$



$$(9384)_{10} = (1101 \ 0111 \ 1100 \ 1000)_{\text{excess 4}}$$

$$(10 \ 0100 \ 1010 \ 1000)_2 \neq (1001 \ 0011 \ 1000 \ 0100)_{BCD}$$

$$\neq (1101 \ 0111 \ 1100 \ 1000)_{\text{excess 4}}$$

[Proved]

[Ans to the question no 2]

$$(7A3.C9)_{16}$$

$$\begin{aligned}(7A3.C9)_{16} &= 7 \cdot 16^2 + 10 \cdot 16^1 + 3 \cdot 16^0 + 12 \cdot 16^{-1} + 9 \cdot 16^{-2} \\&= 1792 + 160 + 3 + 0.75 + 0.03515625 \\&= (1955.78515625)_{10}\end{aligned}$$

$$\begin{array}{r} 5 \overline{) 1955} \\ 5 \overline{) 391} \text{ --- } 0 \\ 5 \overline{) 78} \text{ --- } 1 \\ \quad 5 \overline{) 15} \text{ --- } 3 \\ \quad 5 \overline{) 3} \text{ --- } 0 \\ \quad \quad 0 \text{ --- } 3 \end{array}$$

$$\begin{aligned}0.78515625 \times 5 &= 3.92578125 \\0.92578125 \times 5 &= 4.62890625 \\0.62890625 \times 5 &= 3.14453125 \\0.14453125 \times 5 &= 0.72265625 \\0.72265625 \times 5 &= 3.61328125\end{aligned}$$

$$(1955)_{10} = (30 \ 310)_5$$

$$(1955.78515625)_{10} = (30 \ 310.34303...)_5$$

$$\therefore (7A3.C9)_{16} = (30 \ 310.34303...)_5$$

[Ans to the question no 3]

a)

$$(3A4)_{16} = (0011 \ 1010 \ 0100)_2$$

$\swarrow \quad \downarrow \quad \searrow$
 0011 1010 0100

$$(257)_8 = (010 \ 101 \ 111)_2$$

$\swarrow \quad \downarrow \quad \searrow$
 010 101 111

$$\begin{array}{r}
 (0011 \ 1010 \ 0100)_2 \\
 + (0000 \ 1010 \ 1111)_2 \\
 \hline
 (0100 \ 0101 \ 0011)_2
 \end{array}$$

normal way,

using 2's complement,

$$\begin{array}{r}
 (000000110101)_2 \\
 (111111001010)_{15} \\
 + 1
 \end{array}$$

$$(1111 \ 1100 \ 1011)_{25}$$

$$\begin{array}{r}
 \begin{array}{ccc}
 \overset{1}{0} \overset{1}{2} & \overset{1}{2} \overset{1}{2} & \overset{1}{2} \overset{1}{2} \\
 0100 & 0101 & 0111
 \end{array} \\
 - \begin{array}{ccc}
 11 & 0101 & \\
 \end{array} \\
 \hline
 (0100 \ 0001 \ 1110)_2
 \end{array}$$

$$\begin{array}{r}
 (0100 \ 0101 \ 0011)_2 \\
 + (1111 \ 1100 \ 1011)_2 \\
 \hline
 (10100 \ 0001 \ 1110)_2
 \end{array}$$

↑ discarded

$$\left(\underbrace{0100}_4 \underbrace{0001}_1 \underbrace{1110}_E \right)_2$$

$$(0100 \ 0001 \ 1110)_2 = (41E)_{16}$$

b)

$$\begin{array}{r} 40613 \\ (5433)_7 \overline{) (321654325)_7} \\ \underline{31365} \\ 45564 \\ \underline{0211482} \\ 5433 \\ \underline{226265} \\ 22632 \\ \underline{4333} \end{array}$$

$$(5433)_7 \times 0 = 0$$

$$(5433)_7 \times 1 = (5433)_7$$

$$(5433)_7 \times 2 = (14166)_7$$

$$\begin{array}{r} (5433)_7 \times 2 \\ \hline 14166 \end{array}$$

$$14166$$

$$(5433)_7 \times 3 = (22632)_7$$

$$\begin{array}{r} (5433)_7 \times 3 \\ \hline 22632 \end{array}$$

$$5433_7 \times 4 =$$

$$\begin{array}{r} 5433_7 \times 4 \\ \hline 31365 \end{array}$$

$$\text{So, Quotient} = (40613)_7$$

$$\text{Remainder} = (4333)_7$$

$$\begin{array}{r} 3333 \\ \times 6 \\ \hline 19998 \end{array}$$

$$\begin{array}{r} 5433 \\ \times 6 \\ \hline 32598 \end{array}$$

$$\begin{array}{r} 5433 \\ \times 5 \\ \hline 27165 \end{array}$$

$$\begin{array}{r} 12 \\ \times 12 \\ \hline 144 \end{array}$$

[Ans to the question no 4]

$$(71)_{10}$$

$$= (0100 \ 0111)_2$$

$$= (1011 \ 1000)_{15}$$

+ 1

$$(1011 \ 1001)_{25}$$

$$(63)_{10}$$

$$= (0011 \ 1111)_2$$

$$= (1100 \ 0000)_{15}$$

+ 1

$$(1100 \ 0001)_{25}$$

$$1011 \ 1001$$

$$+ 1100 \ 0001$$

$$\begin{array}{r} 1011 \ 1001 \\ + 1100 \ 0001 \\ \hline 10111 \ 1010 \end{array}$$

discarded

0111 1010 interpreted as signed two's complement
 $= (+122)_{10}$

But, true mathematical sum: $-71 + (-63) = -134$

So, Overflow occurs.

The 8-bit range is -128 to 127

-134 is outside the range, so the result causes signed overflow. The wrapped 8-bit

result $(01101010)_2$ corresponds to $+122$, which is an incorrect answer - evidence of overflow.

[Ans to the question no 5]

Banner costs = $(101101)_2$ dollars $\left| (101101)_2 = 1 \times 2^5 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^0 \right.$
 $= (45)_{10}$ dollars $\left| = (45)_{10} \right.$

Lighting costs = $(57)_8$ Dollars
 $= (47)_{10}$ Dollars

$(57)_8 = 5 \times 8^1 + 7 \times 8^0$
 $= (47)_{10}$

Sound system rental costs = $(2D)_{16}$ dollars
 $= (45)_{10}$ dollars

$(2D)_{16} = 2 \times 16^1 + 13 \times 16^0$
 $= (45)_{10}$

1 good box costs $= (111)_2$ dollars
 $= (7)_{10}$ dollars

$$\therefore 6 \text{ " " " " } = 6 \times 7 = (42)_{10} \text{ dollars}$$

$$\therefore \text{Total cost} = (95)_{10} + (47)_{10} + (45)_{10} + (92)_{10}$$

$$= (179)_{10} \text{ dollars}$$

I have $(1110101)_2$ dollars
 $= 117$ dollars

Inced. $(179 - 117)_{10} = (62)_{10}$ dollars

(Ans)